

Study Engine Visual Additions Report

Locations, teaching purpose, rendered results, and quality checks

3	original diagrams added
302	course pages rebuilt successfully
0	overlap or clipping issues found

Design principles

Every addition is an original responsive SVG that follows the existing Study Engine system: Spirit Purple, warm neutral fills, thin linework, restrained rounded corners, and the established type hierarchy.

The report uses screenshots from the running site, so each example shows the actual student-facing result rather than an isolated design mockup.

Additions at a glance

Course location	New diagram	Problem addressed
Lesson 1.1	Two Python environments	Clarifies the relationship among VS Code, Terminal, Python, libraries, and Colab
Lesson 6.1	Four learning types	Organizes supervised, unsupervised, semi-supervised, and reinforcement learning by signal
Lesson 6.1	ML decision chain	Groups eight steps into framing, data, model, and evidence

Lesson 1.1 - Two Python execution environments

ARCH 498/598
AI for Architecture

Search the course

ALL MODULES
SELECTED READINGS
INTRODUCTION
1 CODING FOUNDATIONS
1.2 Python from zero (I): values, names, and decisions
2 GENERATIVE AI
3 AI-ASSISTED CODING
4 AGENTIC AI
5 EXPLORATORY DATA ANALYSIS
6 MACHINE LEARNING
7 DEEP LEARNING FOUNDATIONS
8 ADVANCED DEEP LEARNING
REFERENCE
COURSE INSTRUCTOR
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MODULE 1 — CODING FOUNDATIONS · 1.1 — SETUP 1: PYTHON, VS CODE, AND GOOGLE COLAB 1 of 8

Before anything else

You have never run a line of Python. That is fine — nobody in this course assumes you have. This guide takes you from a blank laptop to a working setup in about 45 minutes, and every command in it has two versions: one for **Windows**, one for **macOS**. Pick your tab once; the site remembers it on every page.

3 tools to install	~45 min first time, most of it downloading	0 prior programming needed
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Install Python → Install VS Code → Learn the terminal → Install the libraries → Open Google Colab → Verify

Two ways to run the same Python

YOUR LAPTOP

VS Code
edit .py files
see your project

Terminal
run commands
shows the folder

Python
runs the code
uses libraries

Libraries: numpy · pandas · matplotlib · scikit-learn
installed once; imported by your Python program

IN YOUR BROWSER

Google Colab
edit and run notebook cells
Python runs in Google's cloud

Nothing to install
notebooks and a free GPU

PREVIOUS
1.1 — Setup 1: Python, VS Code, and Google Colab

NEXT
Step 1 — Install Python

Location

Coding Foundations / Setup 1 / Before anything else

Why it helps

Students often treat the editor, Terminal, Python interpreter, and libraries as unrelated software. The diagram turns the local route into one continuous chain and places it beside the browser-based Colab route.

Rendered effect

The diagram appears immediately after the installation flow. It establishes the mental model before the detailed setup steps. Purple arrows show execution relationships, while the separate Colab frame distinguishes local and cloud execution.

Study Engine - Visual additions review

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Lesson 6.1 - Four learning types

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SEARCH the course

ALL MODULES

SELECTED READINGS

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1 CODING FOUNDATIONS

2 GENERATIVE AI

3 AI-ASSISTED CODING

4 AGENTIC AI

5 EXPLORATORY DATA ANALYSIS

6 MACHINE LEARNING

6.2 Why statistics and probability matter

6.3 Regression: predicting continuous outcomes

6.4 Classification: predicting categories

6.5 Generalization and reliable model evaluation

7 DEEP LEARNING FOUNDATIONS

8 ADVANCED DEEP LEARNING

REFERENCE

MODULE 6 — MACHINE LEARNING · 6.1 — ML TYPES AND THE COMPLETE WORKFLOW

2 of 5

Four kinds of learning

Supervised	Unsupervised	Semi-supervised	Reinforcement
Examples come with correct answers	No answers — find structure	A few labels, mostly unlabelled	Learns from consequences of actions
Predict energy use from building features	Cluster buildings by consumption pattern	200 labelled facades, 20,000 photos	HVAC control policy over a season

Four kinds of learning: what signal does the model receive?

What accompanies each example?
The answer determines the learning setup

SUPERVISED	UNSUPERVISED	SEMI-SUPERVISED	REINFORCEMENT
A correct answer features + target	No answer features only	A few answers few labels + many examples	Consequences action + reward over time
Predict heating load architecture example regression · classification	Group similar buildings architecture example clustering · representation	Label facade photos architecture example structure, then labels	Control HVAC architecture example policy · reward · feedback

Original course diagram. Standard learning categories, paired with architectural applications used in this course.

PREVIOUS
Central question

NEXT
Where machine learning enters building design

Location

Machine Learning / ML types / Four kinds of learning

Why it helps

The page already had four definition cards, but those cards did not show the shared decision behind the categories. The new root question - what signal accompanies each example - places all four learning types in one logical framework.

Rendered effect

The existing cards remain useful for scanning, while the new diagram explains relationships. Supervised learning uses the warm highlight and purple outline because it is the primary setup used by later course modules.

Lesson 6.1 - Machine-learning decision chain

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Q Search the course

ALL MODULES
SELECTED READINGS
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6 MACHINE LEARNING
Four kinds of learning
Where machine learning enters building design
The complete workflow
Where each module sits
6.2 Why statistics and probability matter
6.3 Regression: predicting continuous outcomes
6.4 Classification: predicting categories
6.5 Generalization and reliable model evaluation
6.6 Demo: UW campus building-energy regression

MODULE 6 — MACHINE LEARNING · 6.1 — ML TYPES AND THE COMPLETE WORKFLOW
4 of 5

The complete workflow

A machine-learning result is a chain of decisions

Leakage or a weak split breaks every step to its right

The model is one stage. The defensible claim is the product.

Original course diagram after the workflow categories synthesized in Lystbæk (2025) and the course's evaluation lessons.

Figure 3. The workflow grouped into framing, data, modelling and evidence. A weak data split or leakage invalidates every later result. Original course diagram after the workflow categories synthesized in Lystbæk (2025).

01
Frame the question
What decision does this inform, at what unit, and what would a useful answer look like?

02
Collect and audit
Check coverage, missing values, duplicates, labelling rules, and who is represented.

PREVIOUS
← Where machine learning enters building design

NEXT
Where each module sits →

Location

Machine Learning / ML types / The complete workflow

Why it helps

Eight sequential steps can look like a software checklist. The diagram groups them into framing, data, model, and evidence, and shows that leakage or a weak split invalidates every result to the right.

Rendered effect

The diagram provides a memorable high-level structure before the detailed workflow. The evidence stage uses the warm highlight to reinforce that the goal is a defensible claim, not merely a trained model.

Overall result and validation

Visual consistency

All three diagrams use the existing Spirit Purple (#4B2E83), warm neutral highlight (#E8E3D3), card radii, and thin arrows. No unrelated visual style was introduced.

Teaching value

The additions answer three recurring questions: where tools execute, how learning types differ, and how a model result becomes evidence. Each topic contains multiple dependent relationships, so a diagram improves comprehension more than another paragraph would.

Copyright and attribution

The diagrams are generated from scratch by the Study Engine code. No slide or paper figure was copied. The workflow uses categories synthesized from Lystbaek (2025), which the page caption identifies.

Technical validation

Check	Result
Python syntax compilation	Passed
Complete course build	Passed - 302 pages / 43 lessons
Desktop browser review	Passed - no overlap, clipping, or unreadable text
Responsive SVG scaling	Passed - viewBox with existing .fig width rules
English-only additions	Passed - no CJK characters in added frontend content
Git whitespace check	Passed

Modified files

Prototype_Phase-01/engine/figures_lit.py

Prototype_Phase-01/content/coding/1.1-setup-python.md

Prototype_Phase-01/content/ml/6.1-ml-types.md

Conclusion: the three diagrams are integrated into the production frontend, improve relationship-based understanding, remain fully in English, and add no unnecessary decoration or copyright risk.